

Dr. D.Y. Patil Unitech Society's
Dr. D.Y. Patil Arts, Commerce and Science College, Pimpri, Pune 18
Department of Computer Science
Academic Year: 2024-25

F.Y.B.Sc.(Data Science) Sem – II
DS-152-P: Lab Course on DS-151-T (Advanced Python Programming)

Practical Assignment3

Date: 20/12/2024

1. Write a function that divides two numbers and handle the case where the divisor is zero.

```
def divide():
    try:
        a=int(input("Enter a:"))
        b=int(input("Enter b:"))
        ans=a/b
        print(ans)
    except Exception as e:
        print("ZeroDivisionError ",e)

divide()
```

2. Write a function that converts a string to an integer and handle cases where the string is not a valid number.

```
def convert():
    try:
        n=int(input("Enter number : "))
    except Exception as e:
        print("string is not a valid number",e)
    else:
        print(n)

convert()
```

3. Write a function that asserts the divisor is zero.

```
def divide():
    a=int(input("enter a:"))
    b=int(input("enter b:"))
    assert b != 0, "Division by zero!"
    return a / b

divide()
```

4. Write a function that asserts the given number is positive.

```
def isPositive():
    n=int(input("Enter number :"))
    assert n<0, "Given Number is Positive"
    print("Given Number is Negative")

isPositive()
```

5. Write a program that reads a list of numbers and asserts that all numbers are positive.
list1 = [0, 10, -20, 30, -40, 50]

```
# Using a for loop to print positive number
for val in list1:
    try:
        assert val < 0, print("Error: Positive")
        print("Negative")
    except Exception as e:
        print(e)
```

6. Write a function that raises this exception if a negative number is passed.

```
class NegativeException(Exception):
    pass
try:
    n=int(input("Enter Number:"))
    if(n <0):
        raise NegativeException ("Error: Negative Number")
    print("Positive Number")
except NegativeException as e:
    print(e)
```

7. Write a program to raise a user defined exception to check if age is less than 18.

```
class AgeError(Exception):
    pass
try:
    age=int(input("Enter Person age:"))
    if(age <18):
        raise AgeError
except AgeError:
    print("Person is not Eligible to vote")
else:
    print("Person is Eligible to vote")
```

8. Write a program to define custom exception as SalaryNotInRangeError. (Salary must be in between 10000 to 50000)

```
class InvalidSalary(Exception):
    def __init__(self):
        print("InvalidSalary: Salary not in range")
    pass
try:
    salary=int(input("Enter a anumber:"))
    if not 10000<salary<50000:
        raise InvalidSalary
    else:
        print("Salary is in range")
except InvalidSalary as e:
    print(e)
```

9. #Python program for FiveDivison Error Exception.

```
class FiveDivisionError(Exception):
    print("Error when Divide a number by five")
    pass
try:
    n1=int(input("Enter Number1:"))
    n2=int(input("Enter Number2:"))
    if n2==5:
        raise FiveDivisionError ("Cannot Divide by Five")
    ans=n1/n2
    print("Answer: ",ans)
except (FiveDivisionError,ZeroDivisionError) as e:
    print(e)
print("End of Program...")
```

OR

```
class FiveDivisionError(Exception):
    def __init__(self):
        print("Error cannot divide by five")
    pass
try:
    n1=int(input("Enter Number1:"))
    n2=int(input("Enter Number2:"))
    if n2==5:
        raise FiveDivisionError
    ans=n1/n2
    print("Answer: ",ans)
except (FiveDivisionError,ZeroDivisionError) as e:
    print(e)
print("End of Program...“)
```